LEAKLISTEN · RISK REGISTER AND HAZARD ANALYSIS

LeakListen risk register and hazard analysis

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
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REVISION HISTORY

Version	Date	Author	Change
0.1	2026-11-01	Amish Chadha	First risk register, preliminary hazard analysis and first-pass design FMEA for option A (stage P3), with the extra hazards of options B and C

LeakListen risk register and hazard analysis

The top project risk is that no utility partner is found, so the cover loss, the pipe materials and the site noise all stay assumptions; the top hazard is work at a street chamber (confined space, traffic and a heavy cover), which the design avoids by placing the logger from the surface; the highest design FMEA scores are the lithium cell failing its first radio pulse after a long rest (RPN 48) and the radio not escaping an iron cover (RPN 36). All scores are the author's judgement, not data. Method: STANDARDS section 23 and ISO 12100. Likelihood, impact, severity and occurrence are scored 1 (low) to 5 (high); detection 1 (certain to be caught) to 5 (unlikely to be caught).

Project risk register

Table 1. Project risks.

#	Risk	Cause	Effect	Likelihood	Impact	Owner (role)	Response
P1	No utility partner	None approached; the first candidate region is the UK (LKL-DEC-001, 2026-10-02)	Reach, cover loss and false alarm rate stay desk estimates; no TRL 4	3	5	Amish	Approach a UK water company with district metered areas before TRL 4
P2	Cover loss too high	Cast-iron or flooded covers block LoRa (20 dB assumed, 30 dB tight)	Nightly summary does not arrive (PR-007)	4	4	Design lead	Gateway within 0.5 km, through-cover antenna where agreed; measure first (LKL-DDR-002 D7)
P3	Sensor data assumed	Ceramic d33, op-amp noise and contact stiffness taken from typical values	Self-noise and bandwidth miss PR-003, PR-004	3	3	Design lead	Check against chosen parts; shaker comparison in CalRig (decided 2026-10-02)
P4	Cost over target	USD 142 against USD 130; main board is a prototype price	Value-engineering target missed (PR-015)	3	2	Amish	Savings in LKL-DEC-001 (fixed-length bar, batch plugs)
P5	Lithium cell supply and shipping	Dangerous goods rules for lithium metal cells	Delays and cost for sample cells	3	2	Design lead	Use a distributor that ships UN3090 cells; LKL-REG-001 G3
P6	Hydrophone variant undefined	Option B is not sized in LKL-CAL-001; hydrant outlets vary	PR-002 (plastic mains) stays open	4	3	Design lead	Size B after Amish chooses (LKL-DDR-004); hydrant survey with the partner
P7	TRL 4 on hold	Portfolio stage cap P3	No hardware evidence	5	3	Amish	Wait for the cap to lift; keep a plan (LKL-PLN-001)

Preliminary hazard analysis

Table 2. Hazards.

#	Hazard	Cause	Harm	Severity	Likelihood	Control	Residual
H1	Confined space entry	Someone enters the chamber to place or fetch a logger	Asphyxia, poisoning	5	2	Placed from the surface by pole; lanyard to recover the logger; no entry without the utility's permit, gas test and standby (LKL-PRC-001 Safety)	2
H2	Heavy cover	Lifting a 30 kg or heavier cover	Back injury, crushed foot	4	3	Lifting key, two people, feet clear	2
H3	Traffic at the chamber	Work in a live street	Death or injury from a vehicle	5	2	Traffic management under local rules (LKL-REG-001 G6)	2
H4	Lithium cell fire or venting	Short, crush, heat or charge of the Li-SOCl2 cell	Burns, toxic gas	5	1	Fuse or PTC, reverse protection, no charging circuit, label (PR-017, PR-018)	2
H5	Magnet pinch and implant interference	The 42 mm pot magnet, about 600 N rated, snaps to steel	Crushed fingers; effect on pacemakers	3	3	Keeper plate on in transport; removed only at the cap; warning in the build plan	2
H6	Dropped neck bar or logger	The bar slips off the neck and falls onto the valve	Damage; falling object	2	3	Lanyard from the bar to the logger; rubber feet wedge the bar; pole placement	1
H7	Pressurised hydrant (option B)	Cap removed or hydrant opened under pressure	A 3.2 kN force on the cap at 10 bar; water hammer; jet	4	2	Not applicable to option A; for B: open and close hydrants slowly, partner's procedure, depressurise before bolting	3
H8	Buried services strike (option C)	Digging a hole and driving a spike near services	Electric shock, gas or water main damage	5	2	Not applicable to option A; for C: services search, hand dig, trained crew	3

Design FMEA (first pass, P3)

Table 3. Failure modes of option A. RPN = severity x occurrence x detection.

#	Item	Function	Failure mode	Effect	Cause	S	O	D	RPN	Action
F1	Lithium cell and capacitor	Supply the radio pulse	First pulse browns out after a long rest	Reset; no uplink the first nights	Passivation of the Li-SOCl2 cell	4	3	4	48	Hybrid layer capacitor (0.059 F needed against 0.1 F fitted, LKL-CAL-001 C3); pulse test at TRL 4 (LKL-STM-001 S9)
F2	Flat antenna	Radio out of the chamber	Cover blocks the signal	No summary received	Cast-iron or flooded cover, 20 to 30 dB	3	4	3	36	Gateway within 0.5 km; measure cover loss first; through-cover antenna where agreed
F3	M12 plug and socket	Sensor link	Corrosion or water in the connector	Noisy or lost signal	Chamber damp; mated for years	3	3	4	36	IP68 A-coded connector; potted overmould; inspect at each visit
F4	Desiccant and seals	Keep the board dry	Condensation in the tube	Corrosion of the board	Humid chamber; pack used up	3	3	4	36	10 g silica gel holds about 2,000 mg against 5 mg of vapour (LKL-CAL-001 J3); O-ring grooves; potting
F5	O-rings and radial screws	Seal the housing	Leak	Water in the logger	Damaged O-ring; screw not seated	4	2	4	32	Compressed O-rings in grooves; the logger floats on its lanyard so it is not submerged for long
F6	Sensor mount	Pass the pipe vibration	Resonance in the band	Wrong spectrum shape	Contact stiffness of a coated cap	2	4	4	32	Calibrate on the shaker (CalRig bay); equaliser (LKL-STM-001 S6)
F7	Pot magnet	Hold the puck	Puck slips off	No signal; puck dangles on its cable	Coating gap over 0.5 mm; non-ferrous cap	3	3	3	27	125 N at 0.5 mm against 100 N; adapter for non-ferrous fittings
F8	Piezo disc and mass	Sense	Bond fails or depolarises	Sensitivity lost	Heat, shock, epoxy ageing	3	2	4	24	Compression mode with a bonded brass mass; ceramic temperature rating to check
F9	Light pipe seal	Show status	Water enters round the rod	Water in the top plug	Epoxy bond cracks	3	2	4	24	Epoxied into a 3 mm hole; blink only at power-up and magnet swipes

#	Item	Function	Failure mode	Effect	Cause	S	O	D	RPN	Action
F10	Neck bar feet	Hold the bar across the neck	Bar slips	Logger and antenna hang on the lanyard	Neck out of the 570 to 710 mm range; smooth wall	2	3	3	18	Rubber pads; locking pin; neck range asked of the partner (LKL-DEC-001)
F11	Real-time clock	Wake in the night window	Drift beyond the window	Listens at the wrong hour	Temperature; missed time sync	2	2	3	12	Weekly time request; 20 s worst (LKL-CAL-001 H2)

Sources

- TFL-RSK-001 of the sister repo TideFlap (format only).
- LKL-CAL-001 for the loads, charges and margins quoted; LKL-STM-001 for the plausible rows.
- Hazards H5 to H8 follow the design notes in LKL-PRC-001 and the option descriptions in LKL-OPT-001; no incident data was consulted.